



Dairy Analytics

Data Analytics Project on Dairy Industry

Strategic Business Intelligence for New Dairy Startups

Using Excel & Tableau for Data-Driven Insights

Presented By :

Amod Sahasrabudhe – B25CE1051

Pratham Shalgaonkar – B25CE1052

Namya Shetty – B25CE1053

Arush Shivpuje – B25CE1054



Introduction

In today's data-driven world, analyzing business data is key to better decisions and efficiency.. The aim is to generate insights that help optimize inventory, reduce waste, and improve business performance.

Project Scope & Goals

The study highlights key insights related to demand, revenue, inventory management, and product shelf life, helping identify opportunities to optimize operations and reduce wastage.

Data cleaning was performed using ChatGPT, followed by analysis in Microsoft Excel using structured sheets and pivot tables. Visualizations were further developed in Tableau to better understand patterns and relationships.





Data Analytics Project

Problem Statement

Dairy businesses face challenges in managing perishable products, maintaining optimal inventory levels, and understanding demand patterns across different locations and sales channels.

Without proper data analysis, this can lead to overstocking, stockouts, and increased product wastage. Therefore, there is a need to analyze dairy data to identify inefficiencies and improve business performance.



Overproduction Issues



Negative Profit Margins



Inconsistent Inventory



High Waste Risk



Objectives

This project aims to leverage data analytics to guide new dairy startups through evidence-based decision making. By analyzing 4,325 transactions across 10 products, 11 brands, and 15 Indian states, we seek to identify optimal strategies for market entry, product selection, and operational efficiency.



**Demand vs
Production
Analysis**



**Revenue &
Profitability**



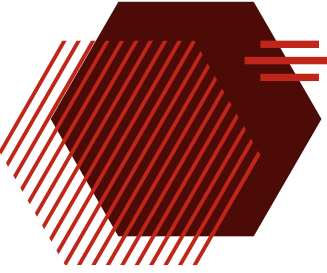
**Inventory Health
Assessment**



**Shelf Life &
Waste Risk**



**Sales Channels
& Geography**



Methodology & Work Steps

Data Foundation & Analysis

Step 1: Data Collection & Understanding – Dataset composition review

Step 2: Data Preparation & Cleaning – Tableau import, calculations

Step 3: Parameter 1 – Demand vs Production Analysis

Step 4: Parameter 2 Analysis – Revenue vs Profitability Analysis

Advanced Parameters & Output

Step 5: Parameter 3 Analysis – Inventory Health Assessment

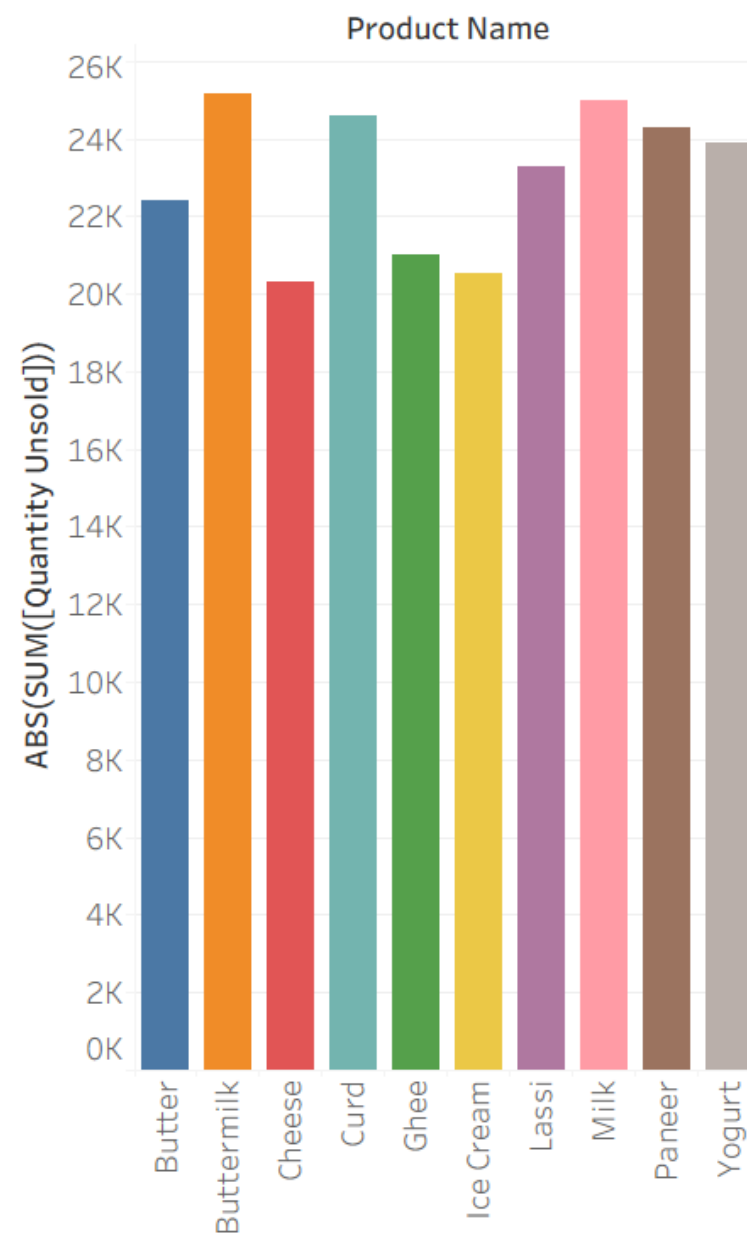
Step 6: Parameter 4 Analysis – Shelf Life & Waste Risk

Step 7: Parameter 5 – Sales Channel & Geography

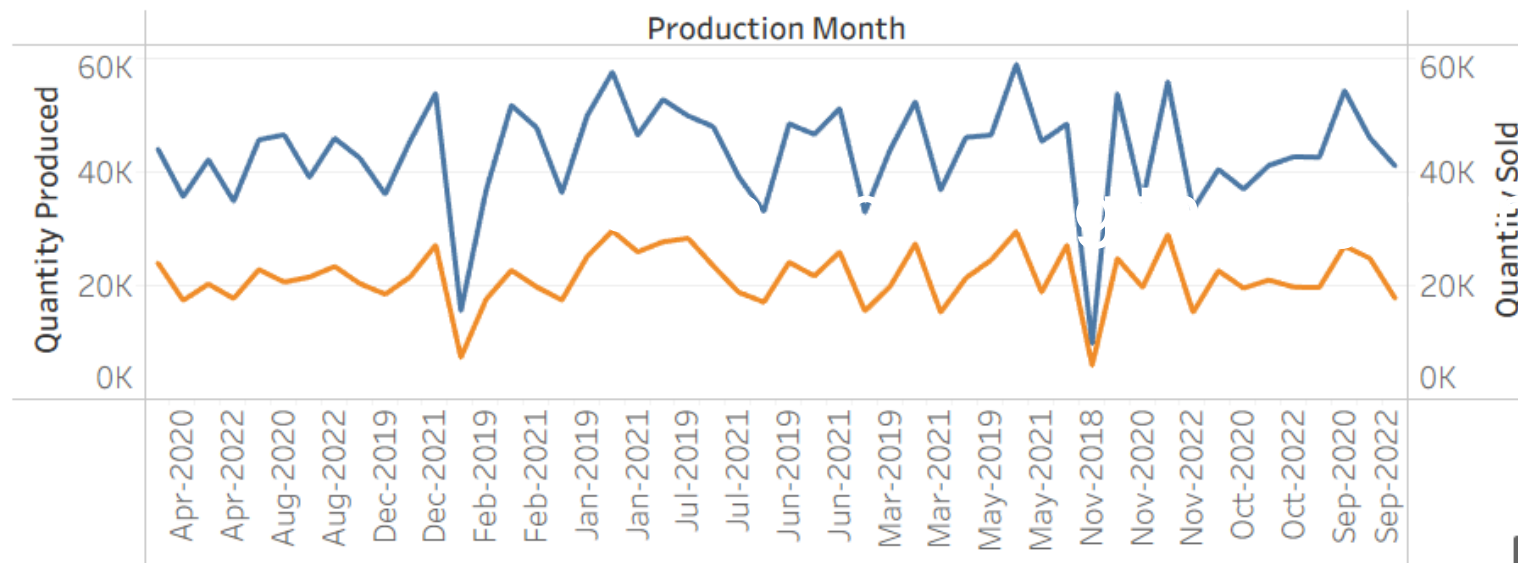
Step 8–9: Integrated Recommendations

Demand vs Production

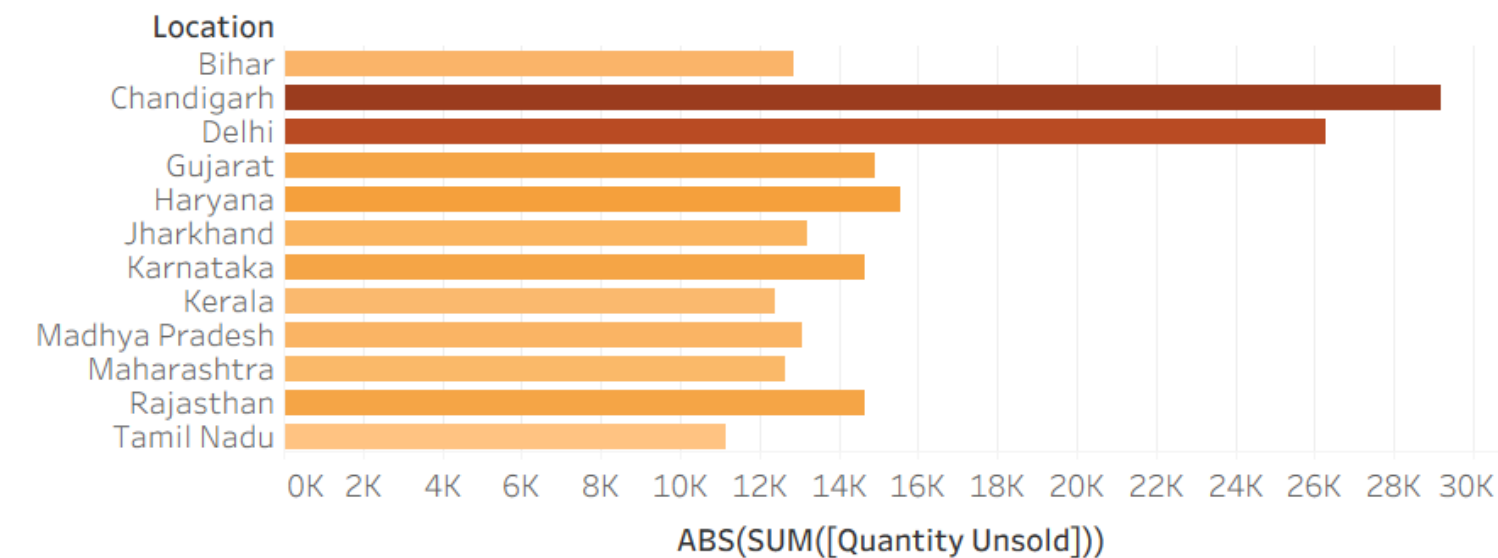
Unsold by product



monthly trend



Unsold by Region



Product Name

- Butter
- Buttermilk
- Cheese
- Curd
- Ghee
- Ice Cream
- Lassi

Measure Names

- Quantity Produced
- Quantity Sold

Quantity Unsold

-29,180 -11,158

Product Name

- ☒ (All)
- ☒ Butter
- ☒ Buttermilk
- ☒ Cheese
- ☒ Curd
- ☒ Ghee

Location

- ☒ (All)
- ☒ Bihar
- ☒ Chandigarh
- ☒ Delhi
- ☒ Gujarat
- ☒ Haryana

Demand vs Production

Are We Producing the Right Amount?

Key Findings:

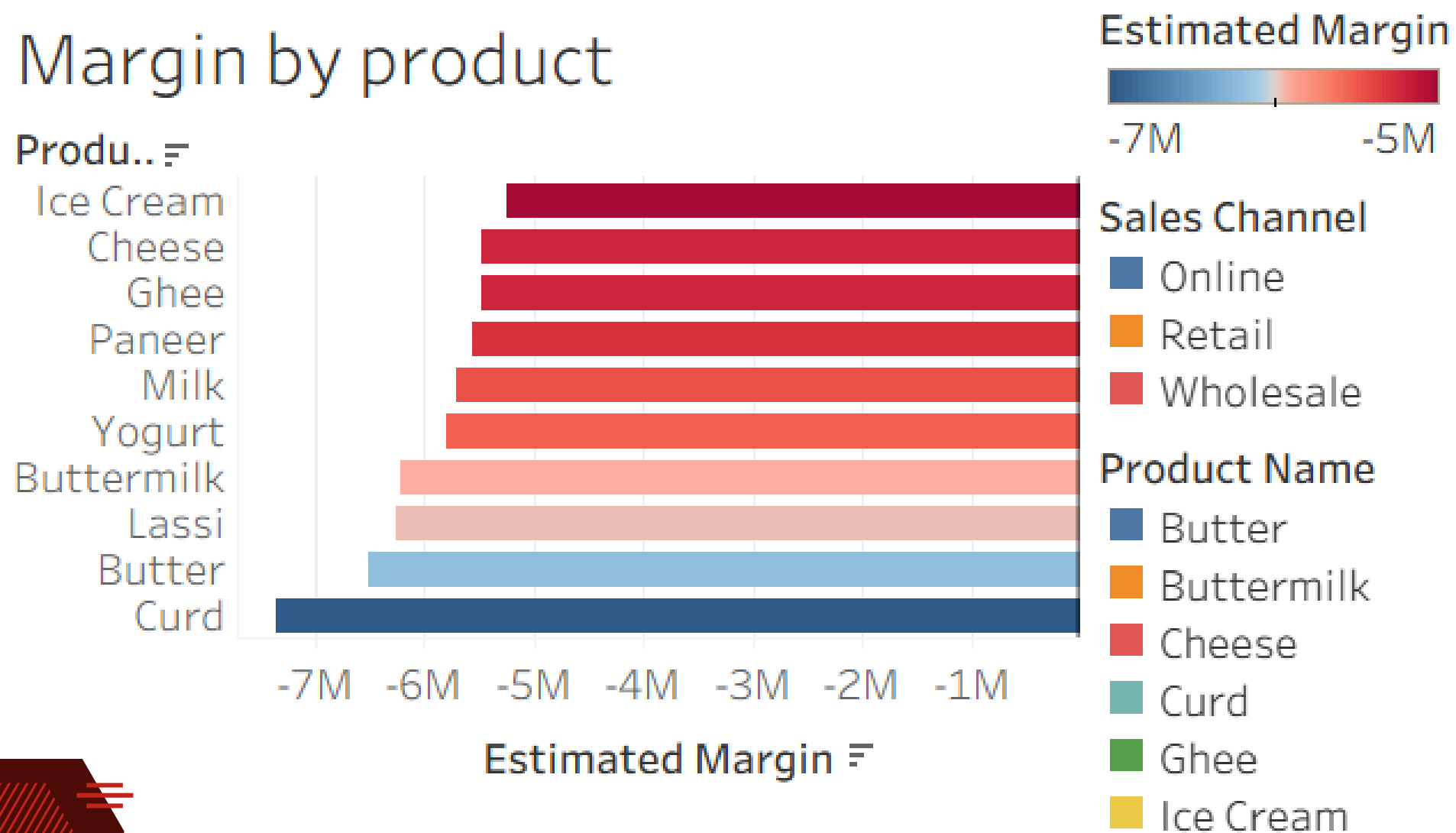
Milk, Buttermilk, and Curd have the highest unsold inventory levels among all products. Chandigarh and Delhi show significantly higher unsold quantities compared to other regions. The monthly trend also reveals noticeable gaps between production and sales, indicating inconsistent demand alignment.

Insight:

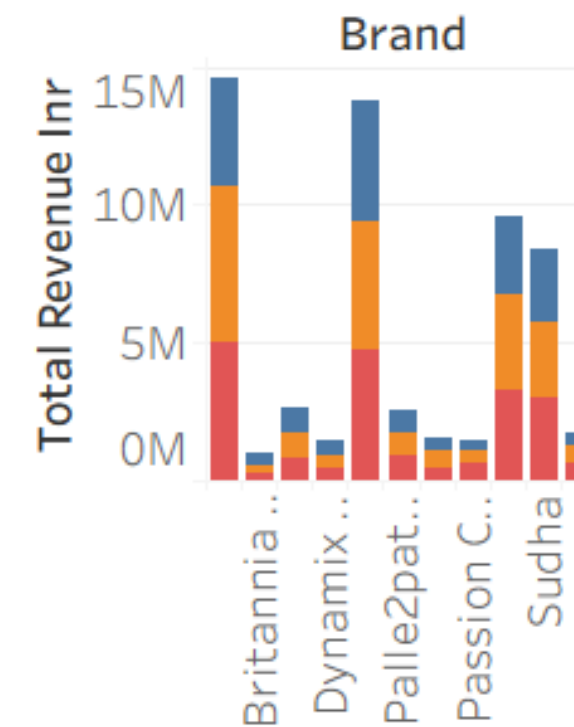
This suggests overproduction in key product categories and specific regions, leading to inventory accumulation. The mismatch between production and actual sales increases the risk of spoilage and waste—especially critical for perishable dairy items with limited shelf life.

Revenue & Profitability Analysis

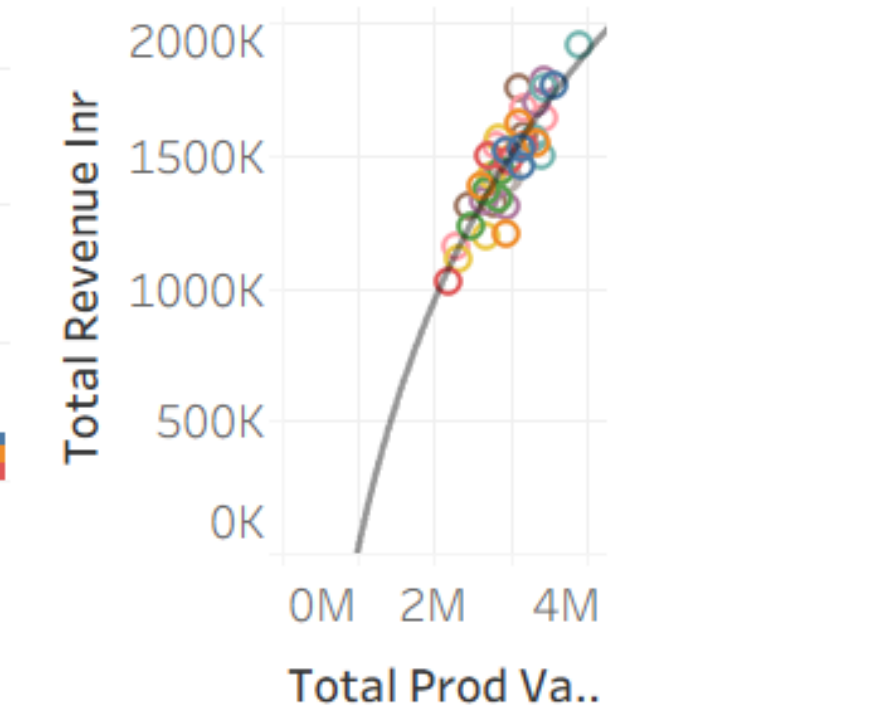
Margin by product



Revenue by channel



Revenue vs Product value



Revenue & Profitability Analysis

Question: What Actually Makes Money?

Key Findings:

Curd and Butter show the lowest (most negative) profit margins, indicating they are the least profitable products. In contrast, Ice Cream and Cheese perform relatively better with higher margins. Retail and Online channels generate higher revenue compared to Wholesale. The revenue vs product value graph shows a strong positive correlation, meaning higher product value directly drives higher revenue.

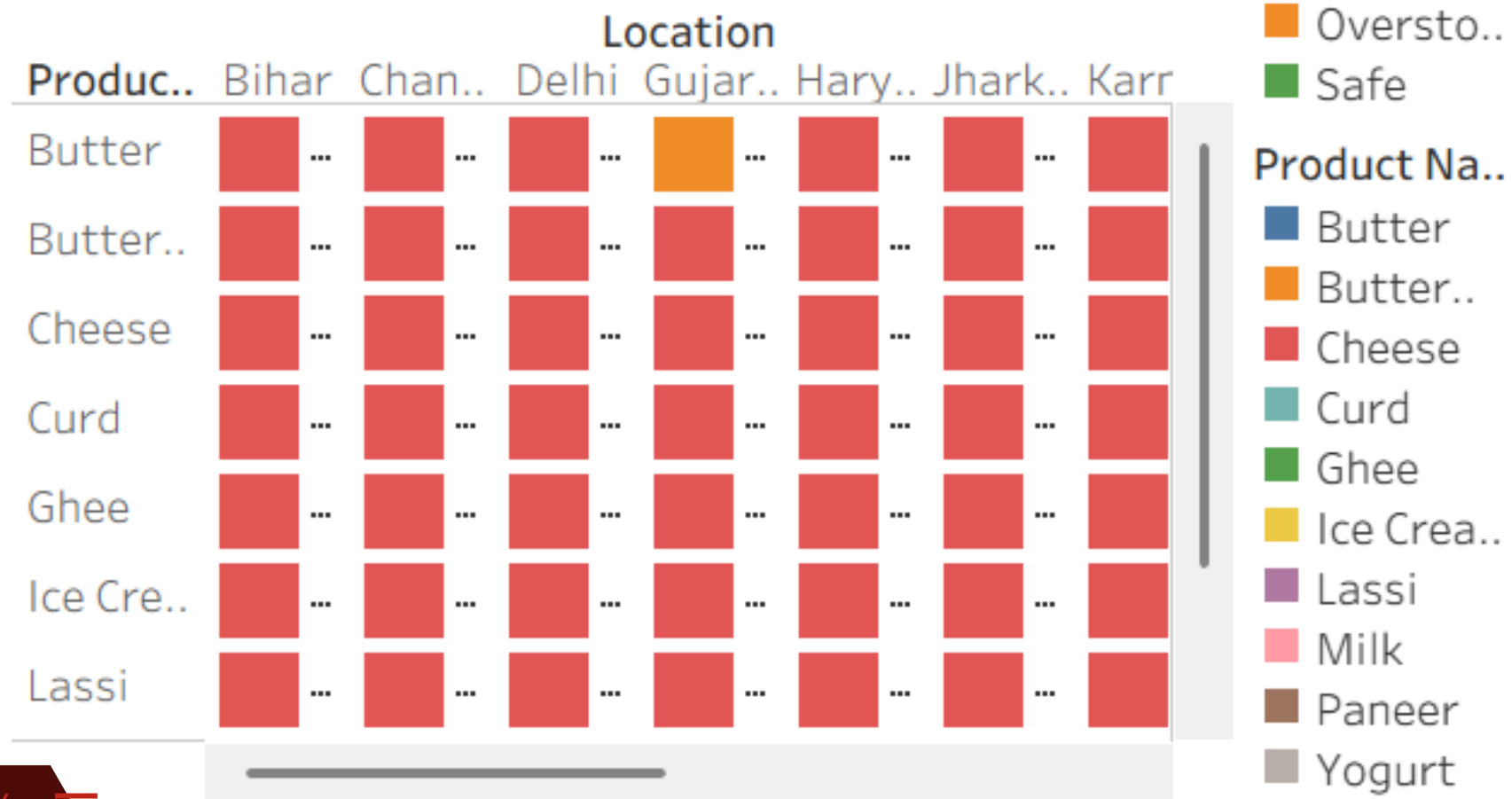
Insight:

This indicates pricing or cost inefficiencies in products like Curd and Butter, leading to consistent losses. At the same time, reliance on Retail and Online channels suggests better profitability opportunities there. Optimizing pricing strategies, reducing production costs, and focusing on high-margin products can significantly improve overall profitability.

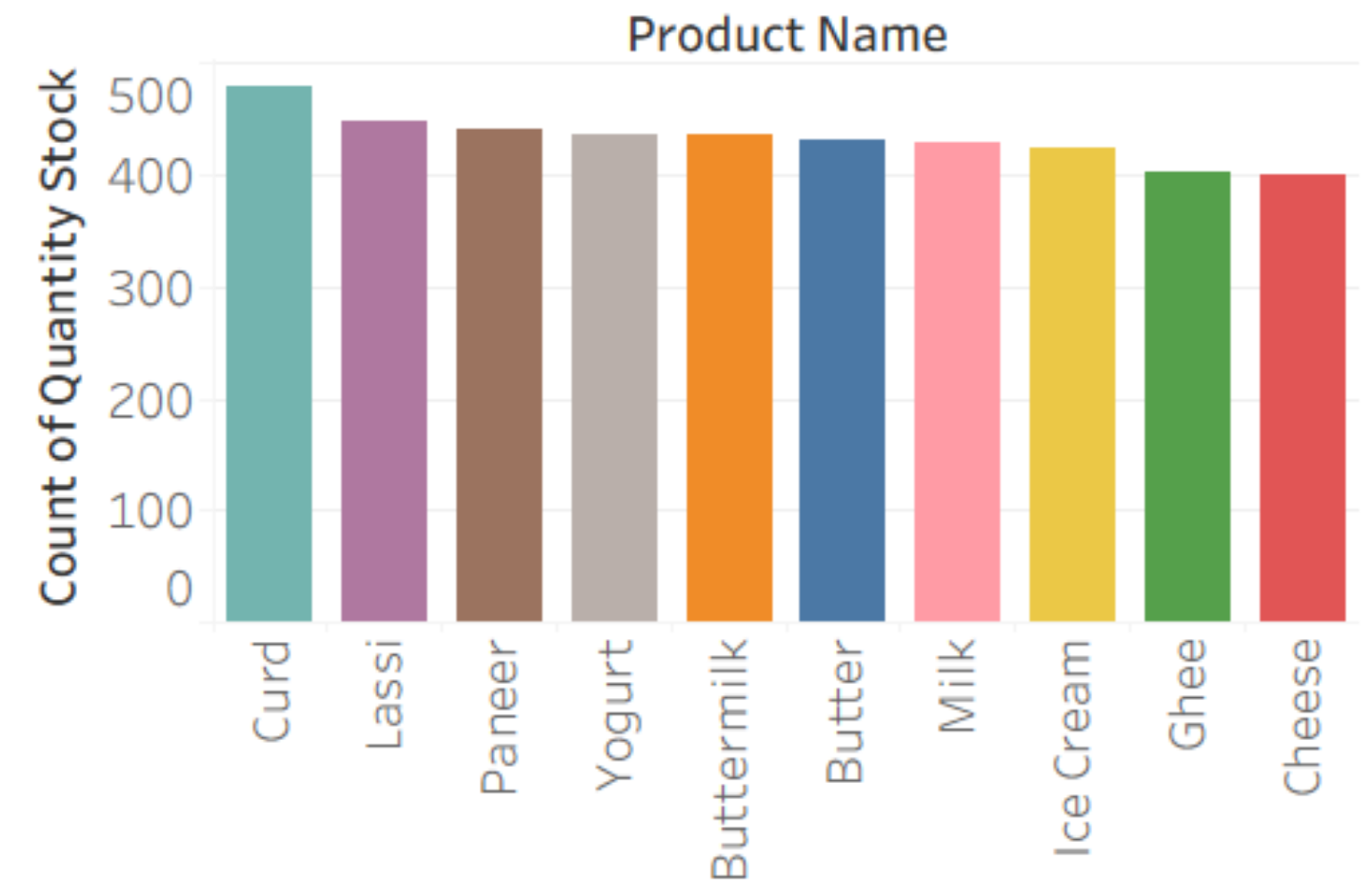


Inventory Health

Stock heatmap



At risk by product





Dairy Analytics

Inventory Health

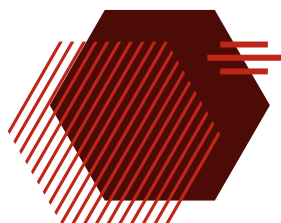
Are We at Risk of Stockouts or Overstock?

Key Findings:

The stock heatmap shows that most products across regions fall under the “At Risk” category, indicating widespread inventory vulnerability. Curd has the highest at-risk stock, followed by Lassi and Paneer, while Cheese and Ghee have relatively lower risk levels.

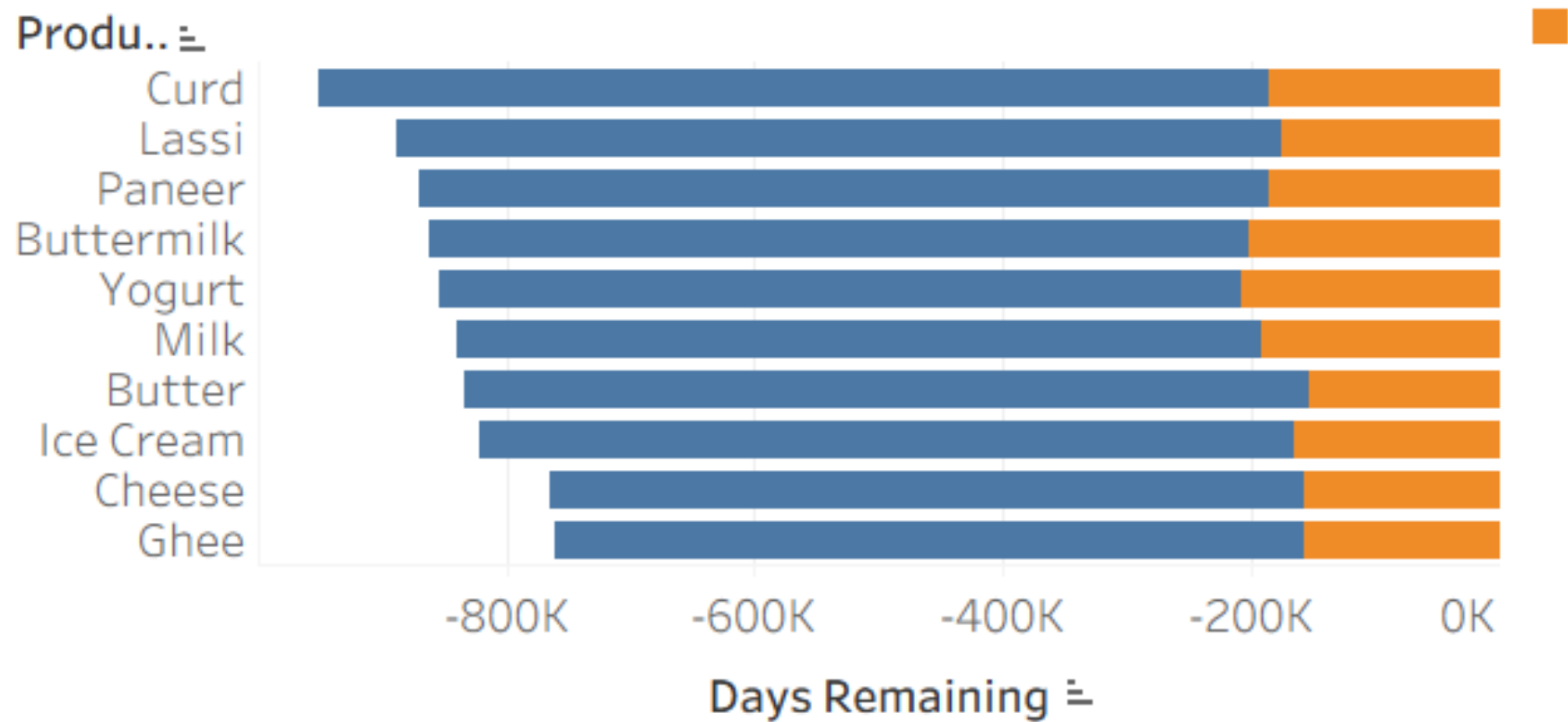
Insight:

This points to poor inventory turnover and inefficient stock management across multiple locations. High at-risk levels—especially for perishable items like Curd—suggest a strong chance of spoilage and waste. Improving demand forecasting, optimizing distribution, and reducing overstocking can help minimize losses and stabilize inventory health.

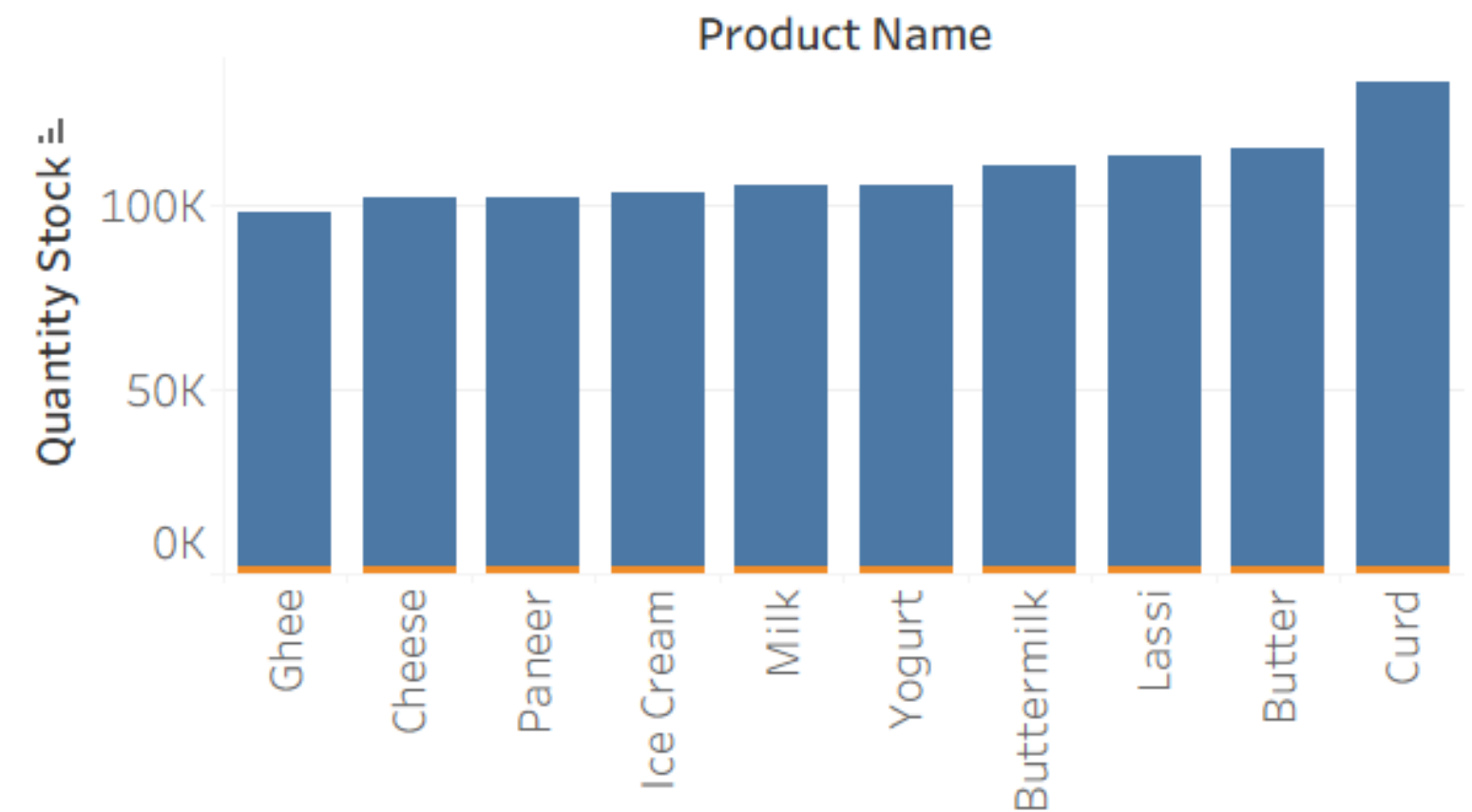


Shelf Life & Waste Risk Analysis

Reorder alert table



Expiry risk by product



Shelf Life & Waste Risk Analysis

Which products are most likely to expire before selling?

Key Findings:

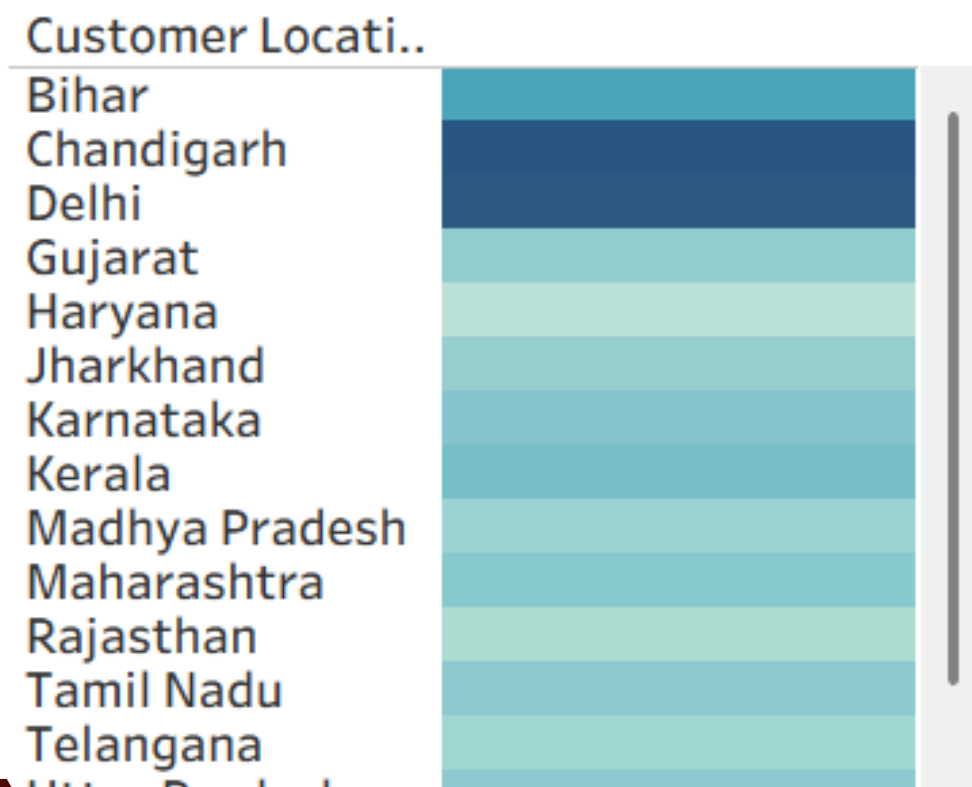
Curd shows the most critical reorder urgency with the lowest days remaining, followed by Lassi and Paneer. The expiry risk chart also highlights Curd as having the highest quantity at risk, with Butter and Lassi close behind. Products like Ghee and Cheese have relatively lower urgency and risk.

Insight:

This indicates a double issue—high expiry risk combined with urgent replenishment needs for certain products, especially Curd. It suggests poor coordination between inventory depletion and restocking cycles. Aligning reorder strategies with actual demand and shelf life can help prevent both stockouts and excessive spoilage.

Sales Channel & Geography

Revenue by Customer region

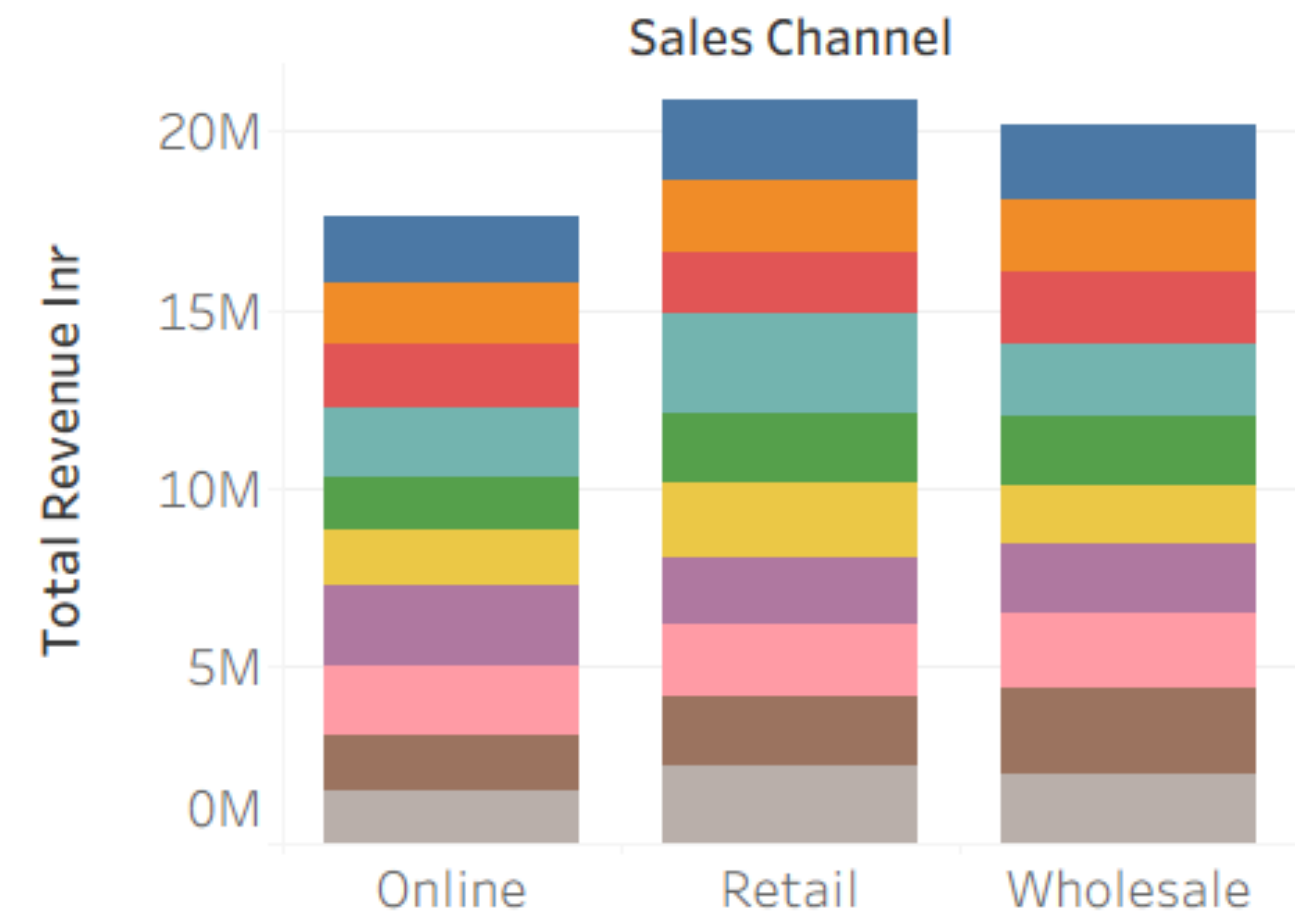


Total Revenue INR
3M  6M

Product Name

- Butter
- Buttermilk
- Cheese
- Curd
- Ghee
- Ice Cream
- Lassi
- Milk
- Paneer
- Yogurt

Revenue by channel-Product



Sales Channel & Geography

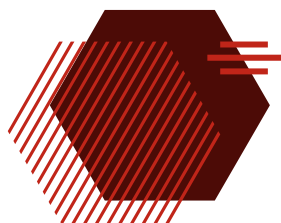
Parameter 5 Analysis

Key Findings:

Delhi and Chandigarh lead in revenue, while Haryana and Rajasthan lag behind. The Retail channel generates the highest revenue, followed by Wholesale, with Online lowest. Core products like Milk, Curd, and Ghee contribute consistently across channels.

Insight:

Revenue is concentrated in a few regions and driven mainly by Retail, indicating limited online penetration and uneven geographic performance. Expanding Online sales and improving reach in lower-performing regions could drive growth.





Integrated Recommendations

Market Entry: Focus on high-revenue regions like Delhi and Chandigarh for strong initial demand

Sales Channel: Prioritize Retail (highest revenue) and expand Online for growth

Product Strategy: Start with Ghee & Butter (low risk), then expand to Paneer, Cheese, and perishables

Inventory: Implement a reorder system to balance stock and reduce waste

Growth Opportunity: Expand into low-performing regions and improve online penetration

Key Insight:

Revenue is concentrated in a few regions and driven mainly by Retail, highlighting the need for geographic and channel expansion.



Delhi or Chandigarh entry



Ghee & Butter portfolio



Retail channel focus



Reorder system critical



Dairy Analytics

Results Summary

Our comprehensive analysis of 4,325 dairy transactions across 10 products and 15 Indian states reveals critical patterns. Overproduction is concentrated in Curd and Buttermilk, while negative margins persist industry-wide. Ghee and Paneer emerge as safer investment options with better profitability potential.



Curd &
Buttermilk
Overproduction



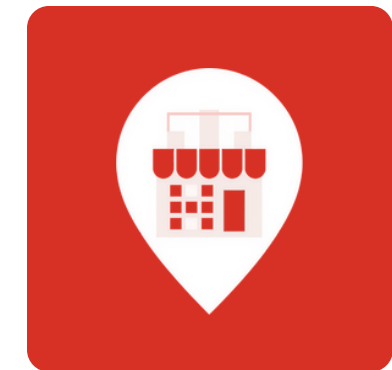
Ghee & Paneer
Safer Bets



Inconsistent
Inventory Control



High Waste Risk
Products



Delhi &
Chandigarh Best
Entry

Conclusions

This project demonstrated how data analytics can effectively guide dairy startup decisions. Through careful product selection, targeted market entry, optimal channel choice, and robust inventory control, new businesses can significantly reduce operational risk and improve revenue potential.

Data-Driven Startup Strategy

Our recommendation to start with Ghee and Butter in Delhi/Maharashtra via retail channels is supported by convergent evidence from multiple parameters. However, pricing and cost challenges require ongoing strategic focus.

THANK YOU

For Your Attention

Class:
CE1

Batch:
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